

# Gabriel Guimarães Carvalho

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## Education

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| <b>Universidade Federal de Pernambuco – UFPE</b><br><i>PhD. Student in Computer Science</i>            | Aug. 2019 – Aug. 2022 (interrupted)<br><i>Recife, BR</i> |
| <b>Università di Roma “La Sapienza”</b><br><i>PhD. degree in Physics and Relativistic Astrophysics</i> | Feb. 2014 – Feb 2017<br><i>Rome, IT</i>                  |
| <b>Universidade Federal de Pernambuco – UFPE</b><br><i>M.Sc. degree in Mathematics</i>                 | 2011 – 2013<br><i>Recife, BR</i>                         |
| <b>Universidade Federal de Pernambuco – UFPE</b><br><i>B.Sc. degree in Mathematics</i>                 | 2008 – 2010<br><i>Recife, BR</i>                         |

## Experience

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| <b>Researcher Scientist</b><br><i>Imec</i> <ul style="list-style-type: none"><li>Computer Vision and Deep Learning researcher</li></ul>                                                                                                                                                                                                              | Jun. 2024 –<br><i>Antwerp, BE</i>                    |
| <b>Senior Researcher Scientist</b><br><i>IDLab - UAntwerpen - Imec</i> <ul style="list-style-type: none"><li>Computer Vision and Deep Learning researcher</li></ul>                                                                                                                                                                                  | Jan. 2024 – May 2024<br><i>Antwerp, BE</i>           |
| <b>Computer Vision Engineer</b><br><i>Moray</i> <ul style="list-style-type: none"><li>Computer Vision and Image Processing</li><li>Data preprocessing and Data Analysis</li><li>Geographic Information System</li><li>Multispectral Drone and Satellite Image Processing</li><li>Pests and diseases detection in agricultural applications</li></ul> | Jul. 2022 – Nov. 2023<br><i>São Paulo/Remote, BR</i> |
| <b>Computer Vision Engineer</b><br><i>SpaceTime Labs</i> <ul style="list-style-type: none"><li>Computer Vision and Image Processing</li><li>Data preprocessing and Data Analysis</li><li>Geographic Information System</li><li>Multispectral Drone and Satellite Image Processing</li></ul>                                                          | Jul. 2021 – Jul. 2022<br><i>São Paulo/Remote, BR</i> |
| <b>Ph.D. Researcher</b><br><i>Projeto CIn - Motorola</i> <ul style="list-style-type: none"><li>Computational Photography</li><li>R&amp;D of image processing and quality assessment tools</li><li>Deep Learning research and development</li><li>Data preprocessing and Data Analysis</li></ul>                                                      | Sep. 2019 – Jun. 2021<br><i>Recife, BR</i>           |
| <b>Postdoctoral Researcher</b><br><i>Universidade Federal Rural de Pernambuco – UFRPE</i> <ul style="list-style-type: none"><li>Alternative gravitational theories and geometric methods in Physics</li></ul>                                                                                                                                        | 2017 – 2018<br><i>Recife, BR</i>                     |
| <b>Lecturer</b><br><i>Universidade Federal de Pernambuco – UFPE</i> <ul style="list-style-type: none"><li>Calculus I, II, III</li><li>Analytic Geometry</li><li>Linear Algebra</li><li>Calculus in one complex variable</li></ul>                                                                                                                    | Nov 2017 – Jul 2019<br><i>Recife, BR</i>             |
| <b>Lecturer</b><br><i>Universidade Federal de Pernambuco – UFPE</i> <ul style="list-style-type: none"><li>Calculus I, II</li><li>Analytic Geometry</li></ul>                                                                                                                                                                                         | Apr 2012 – Jan 2014<br><i>Recife, BR</i>             |

## Technical Skills, Language Skills, and Interests

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**OS:** Linux, macOS, Windows

**Programming Languages:** Python (proficient), C++ (familiar), Java (familiar), Matlab (familiar), Go (beginner)

**Libraries:** OpenCV, Pandas, Matplotlib, Numpy, Scipy, Scikit-Image, Scikit-Learn, PyTorch, PyTorch Lightning, Geopandas, Rasterio

**Version Control:** Git, DVC (familiar)

**Tools:** QGIS, Docker, Lightning Studio

**Cloud:** AWS (familiar)

**Writing:**  $\LaTeX$ , Office

**Languages:** Portuguese (native), English (fluent, C1/C2), Italian (conversational, B2/C1), Norwegian (beginner, A1/A2)

**Interests:** Quantum Computing, Machine Learning, Computer Vision and Image Processing, Mathematics, Physics, Travel, Music, Photography, Nature, Philosophy, Brazilian Jiu-Jitsu

## Extracurricular

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**Projects:** Personal Website and blog posts

**Specialization:** Deep Learning by DeepLearning.AI on Coursera. Certificate earned on July 12, 2020

**Specialization:** Mathematics for Machine Learning by Imperial College London on Coursera. Certificate earned on Dec. 12, 2020

**Course:** Machine Learning by Stanford University on Coursera. Certificate earned on Mar. 18, 2021

**Course:** Introduction to Quantum Information. Certificate earned on Feb. 3, 2025

**Course:** Advanced Computer Vision with TensorFlow. Certificate earned on Feb. 17, 2025